3A 1 THE NATURE OF IONIC BONDING

SPECIFICATION REFERENCE

3.1 PART | 3.2 | 3.3 | 3.5

LEARNING OBJECTIVES

- Describe the formation of ions in terms of loss or gain of electrons.
- Draw dot-and-cross diagrams to show electrons in cations and anions.
- Know that ionic bonding is the result of strong net electrostatic attraction between oppositely charged ions.
- Know and be able to interpret evidence for the existence of ions using electron density maps and from the migrations of ions.

THE FORMATION OF CATIONS AND ANIONS

Some ionic compounds can be formed by the direct combination of two elements.

FORMATION OF SODIUM AND CHLORIDE IONS

For example, sodium chloride can be formed by burning sodium in chlorine:

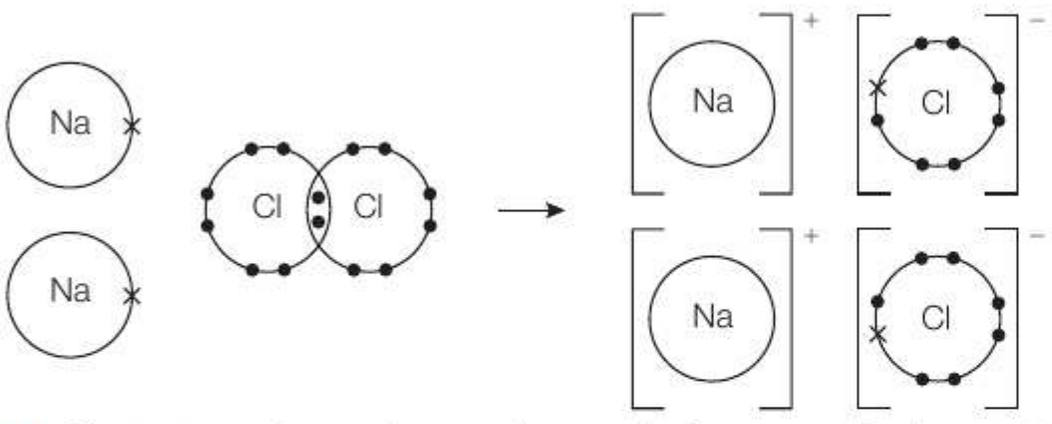
$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

We can represent the reaction that occurs by two ionic half-equations:

$$2Na \rightarrow 2Na^+ + 2e^- \text{ and } Cl_2 + 2e^- \rightarrow 2Cl^-$$

Each sodium atom has lost one electron to become a positive sodium ion. The chlorine molecule has gained two electrons to become two chloride ions.

We can represent the electronic changes involved by dot-and-cross diagrams.



▲ **fig A** Dot-and-cross diagram showing the formation of sodium and chloride ions.

LEARNING TIP

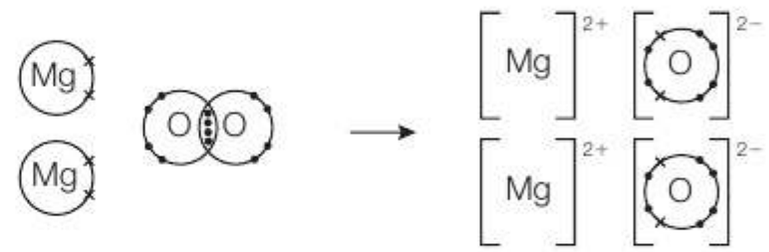
It is important when drawing a dot-and-cross diagram to represent the electrons of one atom using a cross and the other using a dot. You only need to show the outer electrons.

FORMATION OF MAGNESIUM AND OXIDE IONS

Here is the equation for the formation of magnesium oxide:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

A dot-and-cross diagram for the reaction between magnesium and oxygen is shown in **fig B**.

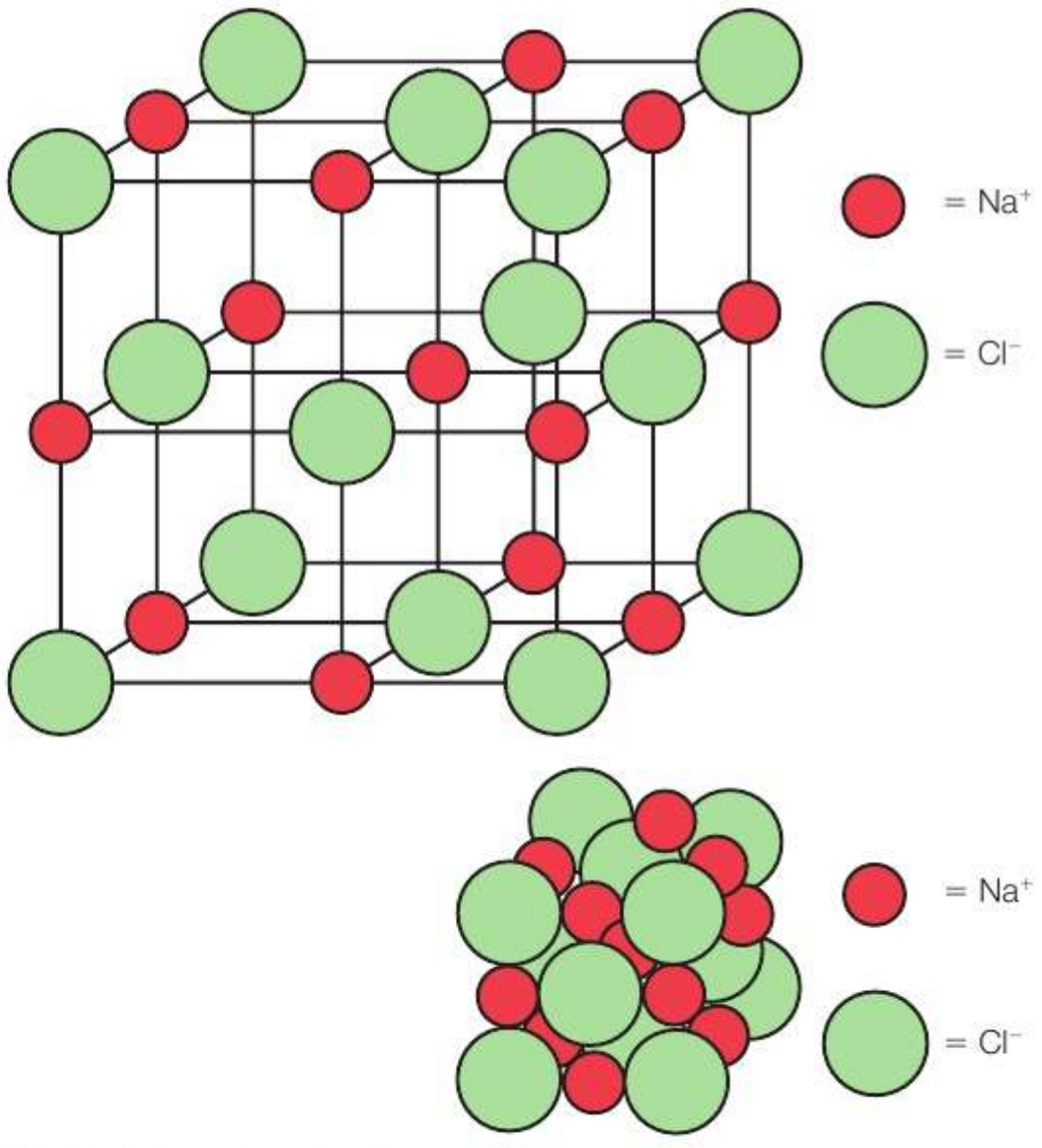


▲ **fig B** Dot-and-cross diagram showing the formation of magnesium and oxide ions.

THE NATURE OF IONIC BONDING

Ionic bonding occurs in solid materials consisting of a regular array of oppositely charged ions extending throughout a giant lattice network.

The most familiar ionic compound is sodium chloride, NaCl. It consists of a regular array of sodium ions, Na^+, and chloride ions, Cl^-, as shown in **fig C**.



▲ **fig C** Structure of sodium chloride.

The diagram on the left is an 'exploded' version of the structure, which is often drawn for the sake of clarity. In practice, the ions are touching one another, as shown in the diagram on the right.

EXAM HINT

It is worth practising drawing a section of an ionic crystal as you may be asked to do so in an exam. Remember to include a key to show which ion is which.

In an ionic solid, there are strong electrostatic interactions between the ions. The ions are arranged in such a way that the electrostatic

attractions between the oppositely charged ions are greater than the electrostatic repulsions between ions with the same charge. The electrostatic interaction between ions is not directional: all that matters is the distance between two ions, not their orientation with respect to one another. (Compare this with covalent bonding in **Topic 3B**.)

When ions are present, the electrostatic interaction between them tends to be dominant. However, it is possible for there to be significant covalent interactions between ions, so you should think of pure ionic bonding as an idealised bonding situation. We will develop this concept further in **Topic 12B (Book 2: IAL)**.

THE STRENGTH OF IONIC BONDING

You can determine the strength of ionic bonding by calculating the amount of energy required in one mole of solid to separate the ions to infinity (i.e. in the gas phase). When they are at an infinite distance from one another, the ions can no longer interact.

Table A below shows the energy required to separate to infinity the ions in one mole of various alkali metal halides.

	AMOUNT OF ENERGY REQUIRED TO SEPARATE THE IONS TO INFINITY / kJ mol^{-1}			
	F^-	Cl^-	Br^-	I^-
Li^+	1031	848	803	759
Na^+	918	780	742	705
K^+	817	711	679	651
Rb^+	783	685	656	628

table A Energy required to break up a lattice of an ionic compound.

For ions of the same charge, the smaller the ions the more energy is required to overcome the electrostatic interactions between the ions and to separate them.

The size of the ions is one factor that affects the strength of ionic bonding, which in turn determines how closely packed the ions are in the lattice.

The lattice energy for lithium fluoride, Li^+F^-, is 1031 kJ mol^{-1}. The equivalent energy for magnesium fluoride, $Mg^{2+}(F^-)_2$, is 2957 kJ mol^{-1}. The radius of the Mg^{2+} ion (0.072 nm) is very similar to the radius of the Li^+ ion (0.074 nm). The increased charge of the Mg^{2+} ion compared to the Li^+ ion results in a significant increase in the strength of the ionic bonding.

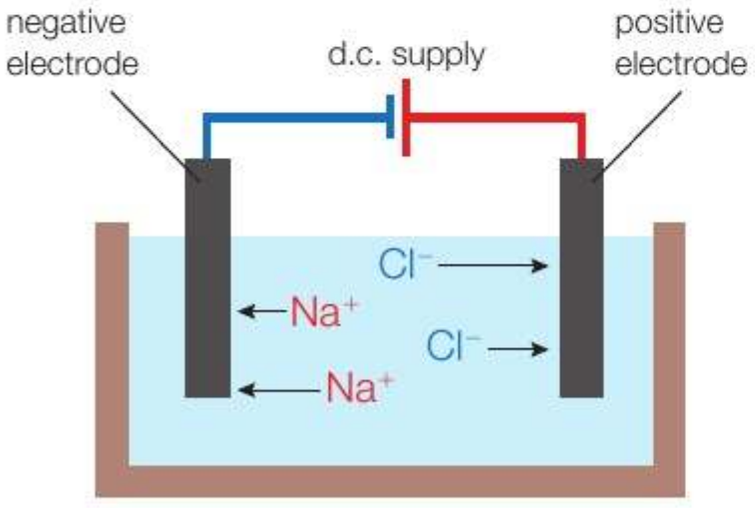
When both cation and anion are doubly charged, the energy required to separate the ions is even larger. For magnesium oxide, $Mg^{2+}O^{2-}$, the value is 3791 kJ mol^{-1}.

There is no simple mathematical relationship to describe the effects that ionic radius and ionic charge have on the strength of ionic bonding. The situation is complicated by the way in which the ions pack together to form the lattice, and by the extent to which there are covalent interactions between the ions. In general, however, the smaller the ions and the larger the charge on the ions, the stronger the ionic bonding.

EVIDENCE FOR THE EXISTENCE OF IONS

Ionic compounds can conduct electricity and undergo electrolysis when either molten or in aqueous solution. This is the most convincing evidence for the existence of ions.

For example, when you pass a direct electric current through molten sodium chloride (**fig D**), sodium is formed at the negative electrode and chlorine is formed at the positive electrode.

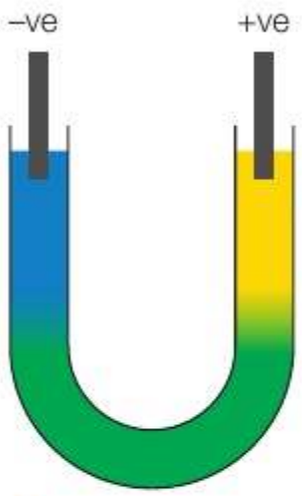


▲ **fig D** Electrolysis of molten sodium chloride.

LEARNING TIP

Avoid saying that there is an 'ionic bond' between two ions.

This is because in an ionic solid, each ion interacts with many other ions, of both the same and opposite charge to itself. The energy that binds the structure does not come from single interactions between ions (so-called ionic bonds), but from the interactions between all of the ions in the lattice. This energy is called the 'lattice energy' and you will meet this in **Topic 12B (Book 2: IAL)**.



▲ **fig E** The effect of passing an electric current through aqueous copper(II) chromate.

The explanation for this phenomenon is that:

- the positive sodium ions migrate towards the negative electrode where they gain electrons and become sodium atoms
- the negative chloride ions migrate towards the positive electrode where they lose electrons and become chlorine molecules.

At the negative electrode: $2Na^+ + 2e^- \rightarrow 2Na$

At the positive electrode: $2Cl^- \rightarrow Cl_2 + 2e^-$

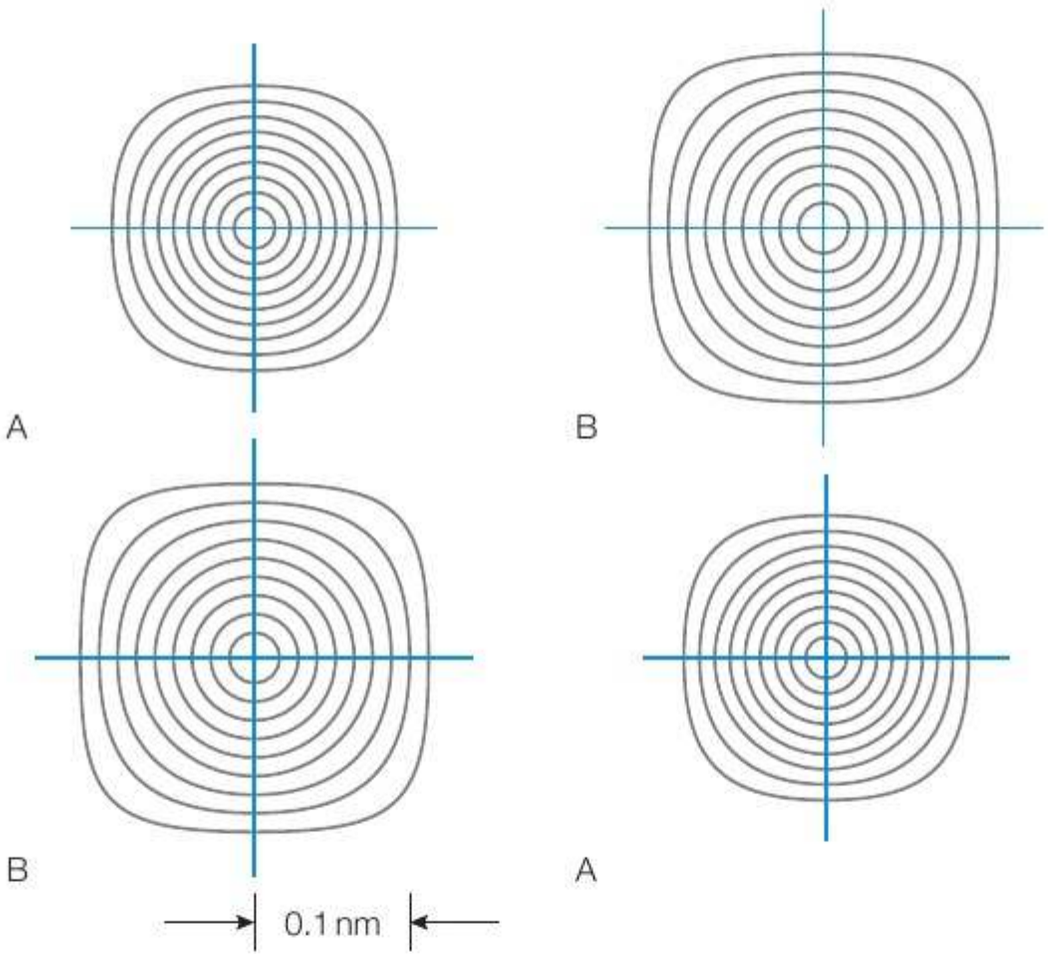
Overall equation: $2NaCl \rightarrow 2Na + Cl_2$

We can demonstrate the movement of ions by passing a direct current through copper(II) chromate(VI) solution (**fig E**). Aqueous copper(II) ions, $Cu^{2+}(aq)$, are blue and aqueous chromate(VI) ions, $CrO_4^{2-}(aq)$, are yellow.

The $Cu^{2+}(aq)$ ions migrate towards the negative electrode and the solution around this electrode turns blue. The $CrO_4^{2-}(aq)$ ions migrate towards the positive terminal and the solution around this electrode turns yellow.

Further evidence for the existence of ions is supplied by electron density maps.

Fig F is an electron density map for sodium chloride.



▲ **fig F** Electron density map of sodium chloride produced from X-ray diffraction patterns.

The electron density map clearly shows separate ions. A represents a sodium ion and B represents a chloride ion.

CHECKPOINT

1. Explain what is meant by the term 'ionic bonding'.

2. Calcium reacts with fluorine to form the ionic compound calcium fluoride:

$$Ca(s) + F_2(g) \rightarrow CaF_2(s)$$

Use a dot-and-cross diagram to show the electronic changes that occur in this reaction.

SKILLS REASONING

3. (a) Suggest why the strength of ionic bonding is greater in sodium fluoride than in potassium fluoride.

(b) Suggest why the strength of ionic bonding in calcium oxide is approximately four times larger than that in potassium fluoride.

SUBJECT VOCABULARY

ionic bonding the electrostatic attraction between oppositely charged ions

3A 2 IONIC RADII AND POLARISATION OF IONS

SPECIFICATION REFERENCE: 3.6 3.7 3.8 3.9

LEARNING OBJECTIVES

- Understand the effects of ionic radius and ionic charge on the strength of ionic bonding.
- Understand reasons for the trends in ionic radii down a group in the Periodic Table, and for a set of isoelectronic ions, e.g. N^{3-} to Al^{3+}.
- Understand the meaning of the term polarisation as applied to ions.

TRENDS IN IONIC RADII

Ionic radii are difficult to measure accurately, and vary according to the environment of the ion. For example, it is important how many oppositely charged ions are touching it (i.e. the co-ordination number). The nature of the ions is also important.

There are several different ways of measuring ionic radii and they all produce slightly different values. If you are going to make reliable comparisons using ionic radii, all the values must come from the same source.

Remember that there are quite large uncertainties when using ionic radii. Trying to explain things in detail is made difficult because of those uncertainties.

GROUP 1		
ION	ELECTRONIC CONFIGURATION	IONIC RADIUS / nm
Li^+	2	0.076
Na^+	2.8	0.102
K^+	2.8.8	0.138
Rb^+	2.8.18.8	0.152

GROUP 7		
ION	ELECTRONIC CONFIGURATION	IONIC RADIUS / nm
F^-	2.8	0.133
Cl^-	2.8.8	0.181
Br^-	2.8.18.8	0.196
I^-	2.8.18.18.8	0.220

table A Trends in ionic radii in Groups 1 and 7.

As you go down each group, the ions have more electron shells; therefore, the ions get larger.

PERIOD 2	N^{3-}	O^{2-}	F^-
Number of protons	7	8	9
Electronic configuration	2.8	2.8	2.8
Ionic radius/nm	0.146	0.140	0.133

PERIOD 3	Na^+	Mg^{2+}	Al^{3+}
Number of protons	11	12	13
Electronic configuration	2.8	2.8	2.8
Ionic radius/nm	0.102	0.072	0.054

table B Trends in ionic radii across a period.

All six of the ions listed in **table B** are isoelectronic. In other words, they have the same number of electrons and therefore the same electronic configuration.

The ionic radius decreases as the number of protons increases.

As the positive charge of the nucleus increases, the electrons are attracted more strongly and are therefore pulled closer to the nucleus.

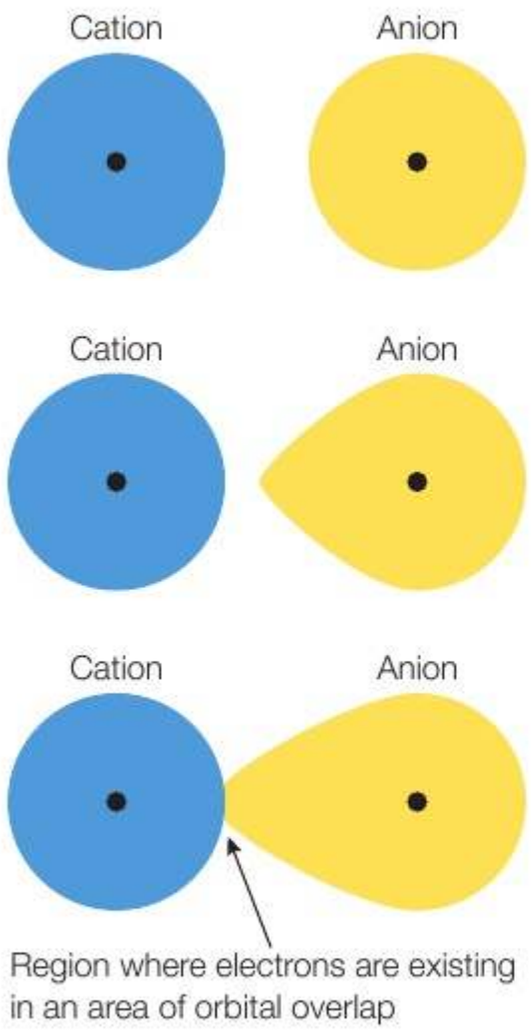
POLARISATION AND POLARISING POWER OF IONS

In an ionic lattice, the positive ion will attract the electrons of the anion. If the electrons are pulled towards the cation, the anion is polarised since the even distribution of its electron density has been distorted.

The extent to which an anion is polarised by a cation depends on several factors. The two main factors are known as Fajan's rules and are summarised here.

DID YOU KNOW?

When you compare chemical values, you should make sure all the data you are comparing come from the same source. For example, the values in tables B and C have all been taken from the Database of Ionic Radii from Imperial College London. However, Chemistry Data Book (JG Stark and HG Wallace) gives a value of 0.171 for the ionic radius of the nitride ion, N^{3-}. In either case, the value is larger than that for the oxide ion, O^{2-}, as expected.



▲ **fig A** A representation of a cation attracting the electrons of an anion in an ionic lattice.

Polarisation will be increased by:

- high charge and small size of the cation (i.e. high charge density of the cation)
- high charge and large size of the anion.

HIGH CHARGE AND SMALL SIZE OF CATIONS

The ability of a cation to attract electrons from the anion towards itself is called its **polarising power**. A cation with a high charge and a small radius has a large polarising power. An approximate value for the polarising power of a cation can be obtained by calculating its charge density. The charge density of a cation is the charge divided by the surface area of the ion. If the ion is assumed to be a sphere, its surface area is equal to $4\pi r^2$, where r is the ionic radius.

An approximation to the charge density can be determined by dividing the charge by the square of its ionic radius.

$$\text{charge density} \sim \frac{\text{charge}}{r^2}$$

HIGH CHARGE AND SMALL SIZE OF ANIONS

The ease with which an anion is polarised depends on its charge and its size. Anions with a large charge and a small size are polarised the most easily.

In an ionic lattice, the polarisation of the anions creates some degree of sharing of electrons between the two nuclei. That is, some degree of covalent bonding exists. You will learn more about this concept in **Topic 12B (Book 2: IAL)**.

CHECKPOINT

1. Explain the trend in the following ionic radii:

(a) $Ca^{2+} > Mg^{2+} > Be^{2+}$

(b) $P^{3-} > S^{2-} > Cl^-$

2. The table gives the ionic radii of some ions.

FORMULA OF ION	IONIC RADIUS / nm
Li^+	0.076
Na^+	0.102
Mg^{2+}	0.072
Al^{3+}	0.054

Arrange the ions in order of their polarising power. Show how you arrived at your answer.

SUBJECT VOCABULARY

polarising power the ability of a positive ion (cation) to distort the electron density of a neighbouring negative ion (anion)

polarisation the distortion of the electron density of a negative ion (anion)

3A 3 PHYSICAL PROPERTIES OF IONIC COMPOUNDS

SPECIFICATION REFERENCE
3.1 PART 3.4

LEARNING OBJECTIVES

- Be able to explain the physical properties of ionic compounds in terms of their bonding and structure.

Ionic compounds typically have the following physical properties:
- high melting temperatures
- brittleness
- poor electrical conductivity when solid but good when molten
- often soluble in water.

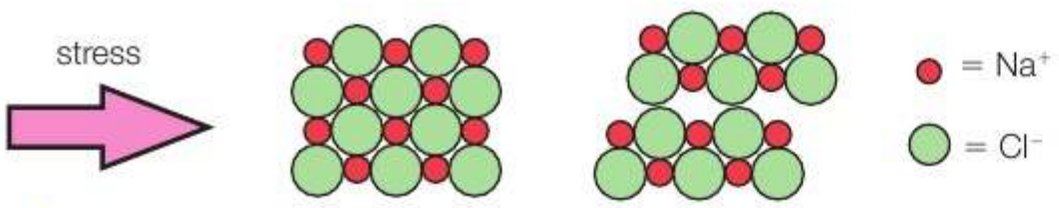
HIGH MELTING TEMPERATURES

Ionic solids consist of a giant lattice network of oppositely charged ions (see **Topic 2B.2**). There are many ions in the lattice and the combined electrostatic forces of attraction among all of the ions is large.

A large amount of energy is required to overcome the forces of attraction sufficiently for the ions to break free from the lattice and slide past one another.

BRITTLENESS

If a stress is applied to a crystal of an ionic solid, then the layers of ions may slide over one another.



▲ **fig A** Effect of stress on an ionic crystal.

Ions of the same charge are now side by side and repel one another. The crystals break apart.

ELECTRICAL CONDUCTIVITY

Solid ionic compounds do not, in general, conduct electricity. This is because there are no delocalised electrons and the ions are also not free to move under the influence of an applied potential difference.

However, molten ionic compounds *will* conduct since the ions are now mobile and will migrate to the electrodes of opposite sign when a potential difference is applied. If direct current is used, the compound will undergo electrolysis as the ions are discharged at the electrodes.

EXAM HINT

Remember, oxidation always takes place at the anode.

Aqueous solutions of ionic compounds also conduct electricity and undergo electrolysis, since the lattice breaks down into separate ions when the compound dissolves.

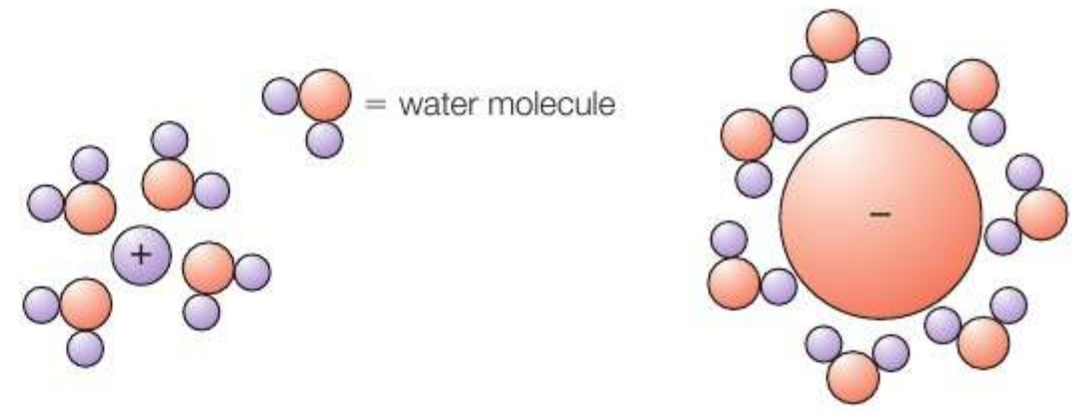
DID YOU KNOW?

As nearly always in chemistry, there are exceptions. Some ionic compounds do conduct electricity when solid. For example, solid lithium nitride (Li_3N) will conduct electricity and is used in batteries for this reason.

SOLUBILITY

Many ionic compounds are soluble in water. We will explain this solubility more fully in **Topic 12A (Book 2: IAL)**, including the part played by entropy changes.

At the moment, you just need to understand that the energy required to break apart the lattice structure and separate the ions can, in some instances, be supplied by the **hydration** of the separated ions produced. Both positive and negative ions are attracted to water molecules because of the polarity that water molecules possess (see **Topic 3B** for an explanation of polarity).



The oxygen ends of the water molecules are attracted to positive ions.

The hydrogen ends of the water molecules are attracted to negative ions.

▲ **fig B** Hydration of ions.

CHECKPOINT

1. Explain why sodium chloride:
(a) does not conduct electricity when solid, but does when molten
(b) has a high melting temperature
(c) is soluble in water.

SUBJECT VOCABULARY

hydration the process of water molecules being attracted to ions in solution and surrounding the ions; the oxygen ends of the water molecules are attracted to the positive ions (cations); the hydrogen ends of the water molecules are attracted to the negative ions (anions); hydration of ions is an exothermic process (i.e. heat energy is released)